

SoftwareVersion.

The inputs to the conceptual algorithm are:

- A subset of the fields of the [QueryImage](#) command as a structure named **requestor**:
- **VendorID** of the requestor as **vendorID**
- **ProductID** of the requestor as **productID**
- Current **SoftwareVersion** of the requestor as **softwareVersion**
- The list of all current entries for the given VendorID and ProductID from the **DeviceSoftwareVersionModel** schema, ordered by SoftwareVersion, accounting for the following fields, as an array **candidates[]**
 - **SoftwareVersion** of the entry as **softwareVersion**
 - **SoftwareVersionValid** of the entry as **softwareVersionValid**
 - **MinApplicableSoftwareVersion** of the entry as **minApplicableSoftwareVersion**
 - **MaxApplicableSoftwareVersion** of the entry as **maxApplicableSoftwareVersion**

The output of the conceptual algorithm is a tuple of:

- **candidateWasFound**, a boolean predicate indicating whether a newer version candidate was found
- **softwareVersionFound**, the version of the newer version candidate, if **candidateWasFound** was true, 0 otherwise.

The algorithm is as follows:

1. Assume no matching candidate found
 - **candidateWasFound** := False
 - **softwareVersionFound** := 0
2. Obtain candidates from a DCL replica or from a DCL-based dataset for the given **vendorID** and **productID** in the query, keeping only entries where **softwareVersionValid** is true.
 - An OTA Provider MAY as well filter available versions by certification compliance status (see [Section 11.23.8, “DeviceSoftwareCompliance / Compliance test result Schema”](#)).
3. Sort all candidates by ascending **softwareVersion**.
4. Iterate through all candidates to find all positive matches within the sorted candidates. A "positive match" is a candidate which fullfills every condition in the following list:
 - **requestor.softwareVersion** < **candidate.softwareVersion**
 - **requestor.softwareVersion** ≥ **candidate.minApplicableSoftwareVersion**
 - **requestor.softwareVersion** ≤ **candidate.maxApplicableSoftwareVersion**
5. From the positive matches, select the very last one of list, which will be the newest (numerically highest) possible softwareVersion that could be used. If no positive matches were found, no new software version is available.